

COMPASS_SynopticCB_PW_DOC_202507_Data

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##Setup - Change things here & write any notes

```
#identify section  
cat("Setup Information")
```

Setup Information

```
##### Run information - PLEASE CHANGE  
Date_Run = "7/27/25" #Date that instrument was run  
Run_by = "Stephanie J. Wilson" #Instrument user  
Script_run_by = "Stephanie J. Wilson" #Code user  
run_notes = "Some checks off due to faulty second standard curve." #any notes from the run  
samples <- c("GCW", "GWI", "MSM", "SWH") #whatever identifies your samples within the same names  
samples_pattern <- paste(samples, collapse = "|")  
  #samples_pattern <- "GCW" #use this instead of the line above if you have only one site code  
chks_name = "Chk_Std_50C2N" #what did you name your check standards?  
  
##### File Names - PLEASE CHANGE  
#file path and name for raw summary data file  
raw_file_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202507.txt"  
  
#file path and name for raw all peaks file  
raw_allpeaks_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202507_allpeaks.txt"  
  
#file path and name of processed data file  
processed_file_name = "Processed Data/COMPASS_SynopticCB_PW_Processed_DOC_202507.csv"  
  
##### Log Files - PLEASE CHECK  
#downloaded metadata csv - downloaded from Google drive as csv for this year  
Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2025.csv"  
  
#qac log file path for this year  
Log_path = "Raw Data/COMPASS_Synoptic_TOCTN_QAClog_2025.csv"
```

##Set Constants Up Code

##Read in metadata and create similar sample IDs for matching to samples

Import Data Functions

Import Sample Data

```
## Import Sample Data

## New names:
## * `` -> '...14'

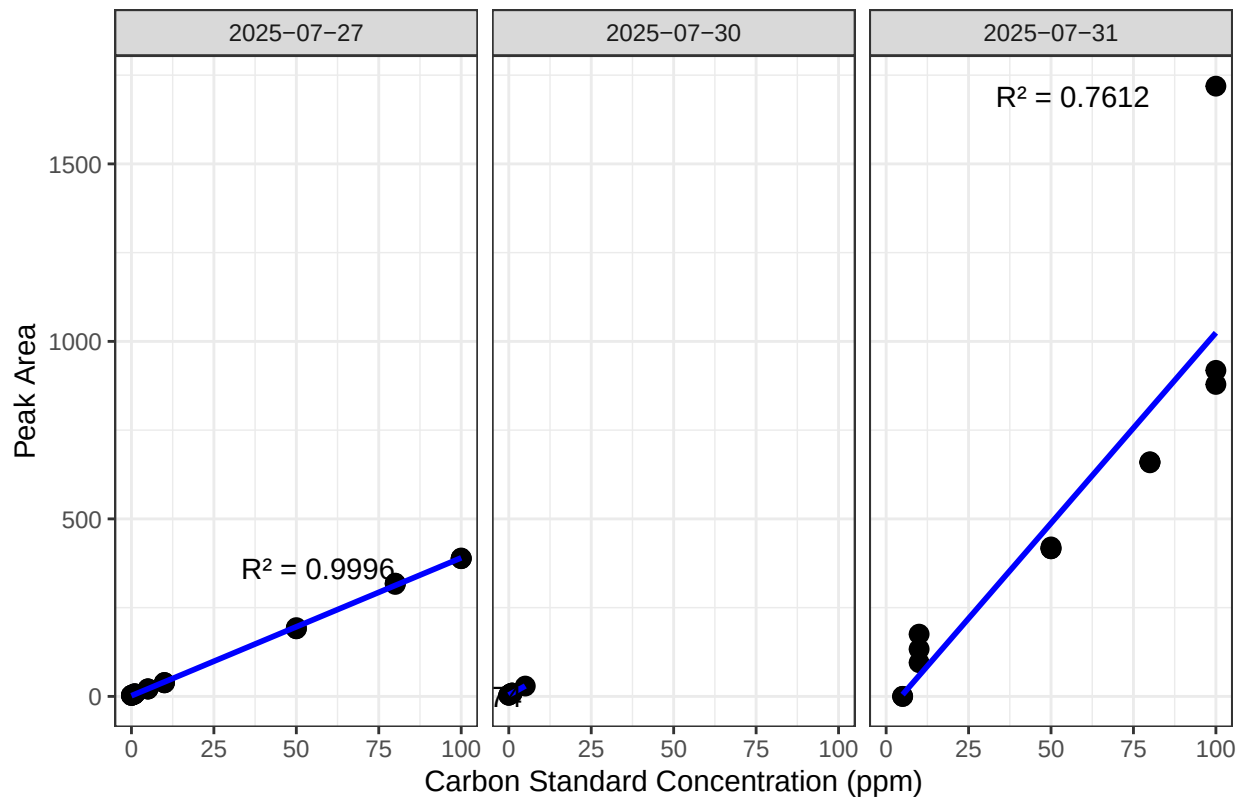
## # A tibble: 6 x 4
##   sample_name      npoc_raw tdn_raw run_datetime
##   <chr>          <dbl>   <dbl> <chr>
## 1 202507_GCW_UP_LysB_20cm    20.6    0.587 7/27/2025 10:40:38 PM
## 2 202507_GCW_TR_LysA_20cm     7.35    0.297 7/27/2025 11:08:12 PM
## 3 202507_GCW_TR_LysB_10cm     6.76    0.542 7/27/2025 11:37:58 PM
## 4 202507_GCW_TR_LysB_20cm     9.30    0.456 7/27/2025 11:55:48 PM
## 5 202507_GCW_TR_LysC_10cm    15.7    0.313 7/28/2025 12:24:26 AM
## 6 202507_GCW_TR_LysC_20cm     4.18    0.147 7/28/2025 12:48:05 AM
```

Assessing Standard Curves

```
## Assess the Standard Curves

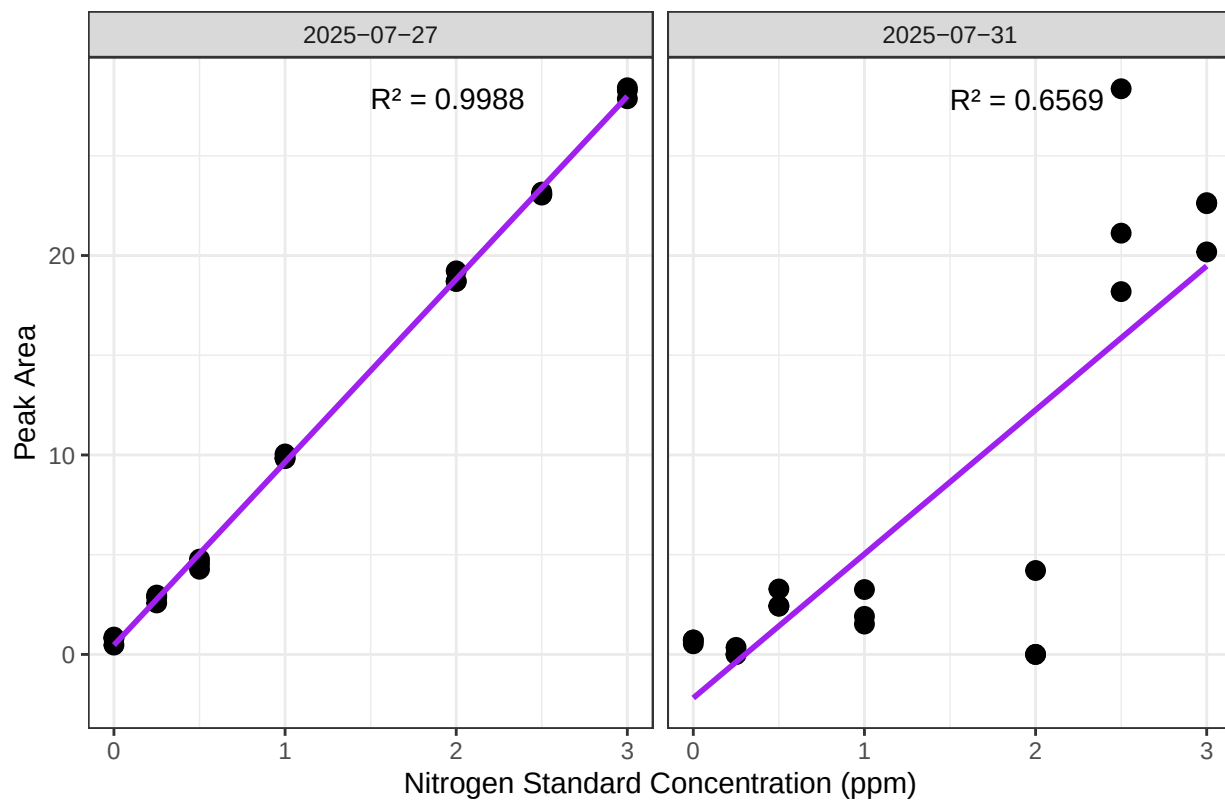
## New names:
## 'geom_smooth()' using formula = 'y ~ x'
## * `` -> '...18'
```

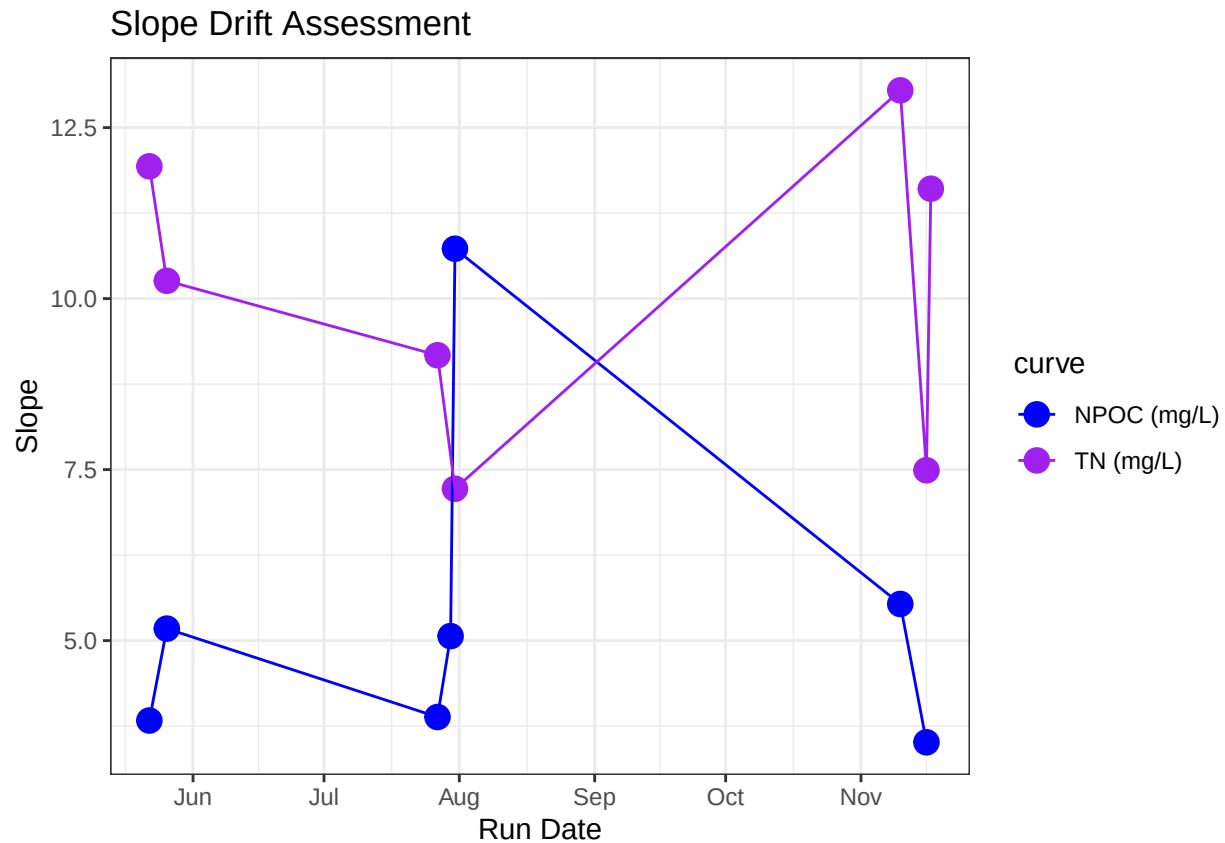
NPOC Std Curve by Date



```
## 'geom_smooth()' using formula = 'y ~ x'
```

TN Std Curve by Date





```
## [1] "NPOC Curve r2 GOOD"
```

```
## [1] "TN Curve r2 GOOD"
```

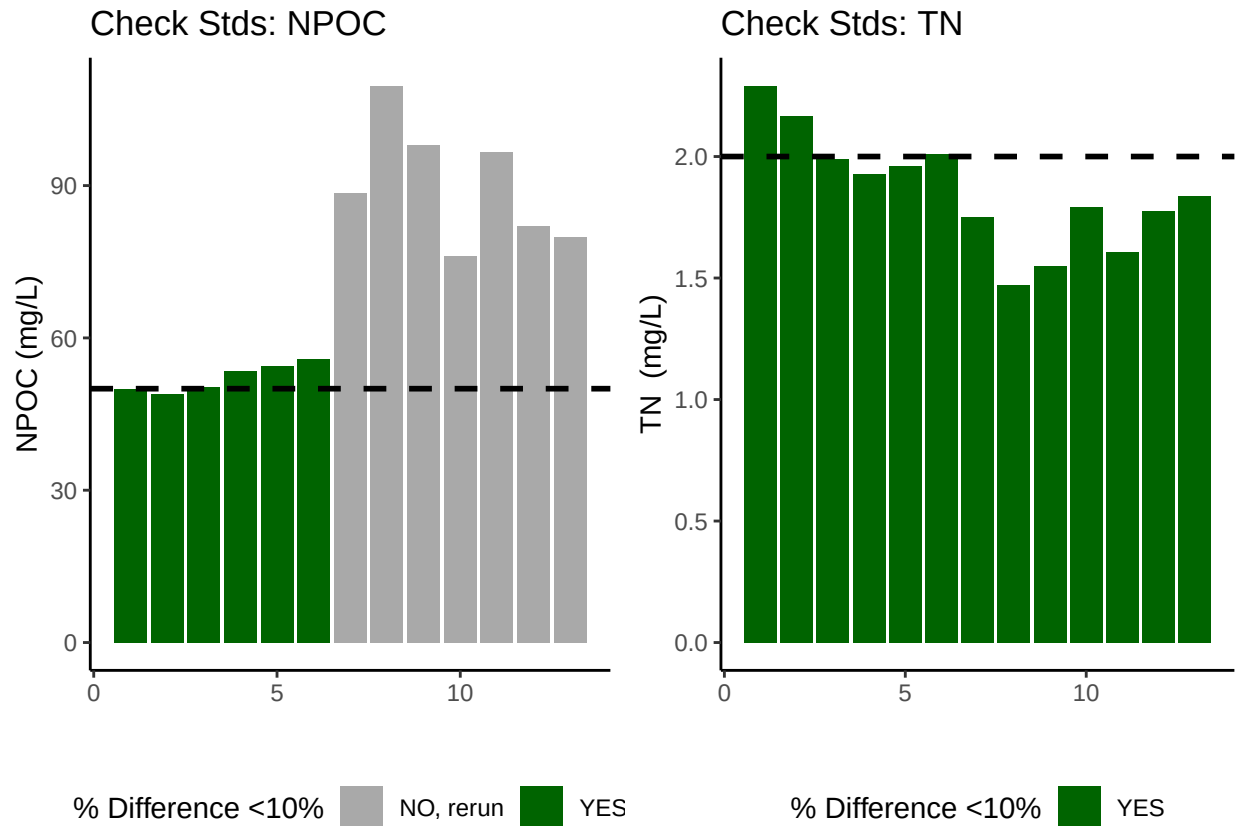
Assess Check Standards

```
## Assess the Check Standards
```

```
## New names:
## * ' ' -> '...14'
```

```
## [1] "Carbon CHECK STANDARD RSD TOO HIGH - REASSESS"
```

```
## [1] "Nitrogen CHECK STANDARD RSD TOO HIGH - REASSESS"
```



```
## [1] ">60% of Carbon Check Standards are within range of expected concentration"
```

```
## [1] ">60% of Nitrogen Check Standards are within range of expected concentration"
```

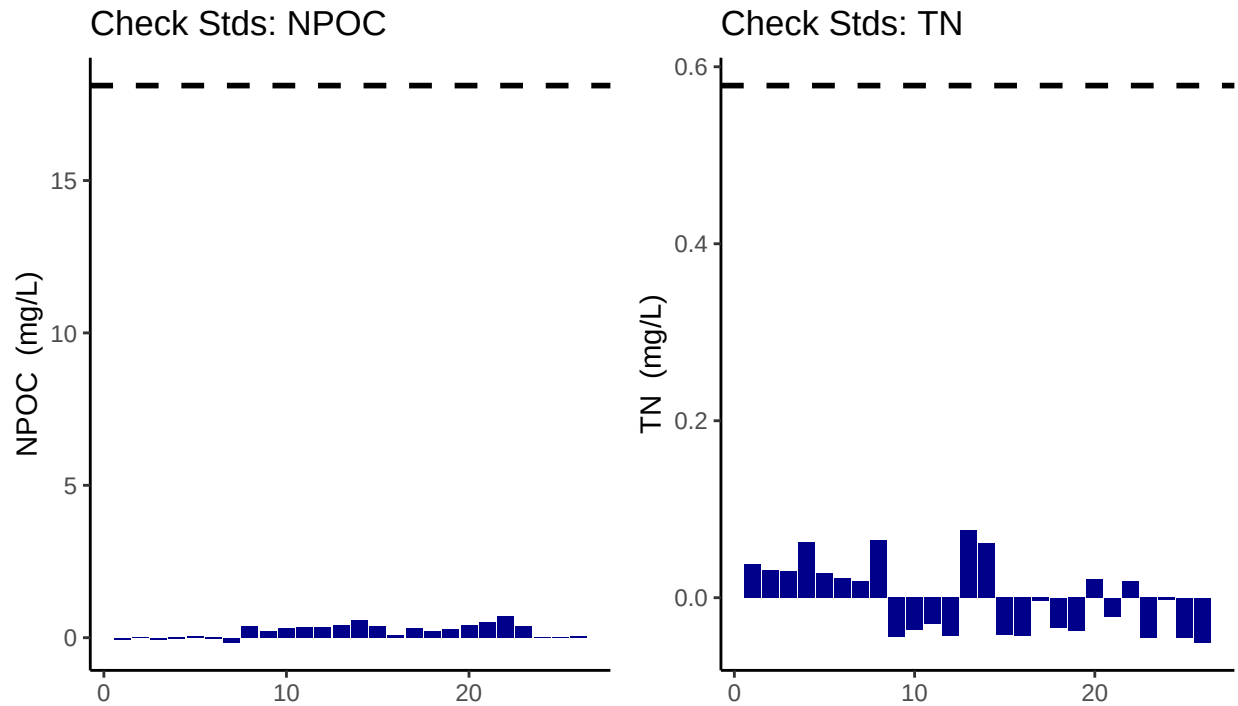
Assess Blanks

```
## Assess Blanks
```

```
## New names:
## * ' ' -> '...14'
```

```
## [1] ">60% of Carbon Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">60% of Nitrogen Blank concentrations are lower 25% quartile of samples"
```



Blank Conc <25% Quartile Samples ■ YE Blank Conc <25% Quartile Samples ■ YE

```
## carbon blanks:
```

```
## [1] 0.2175877
```

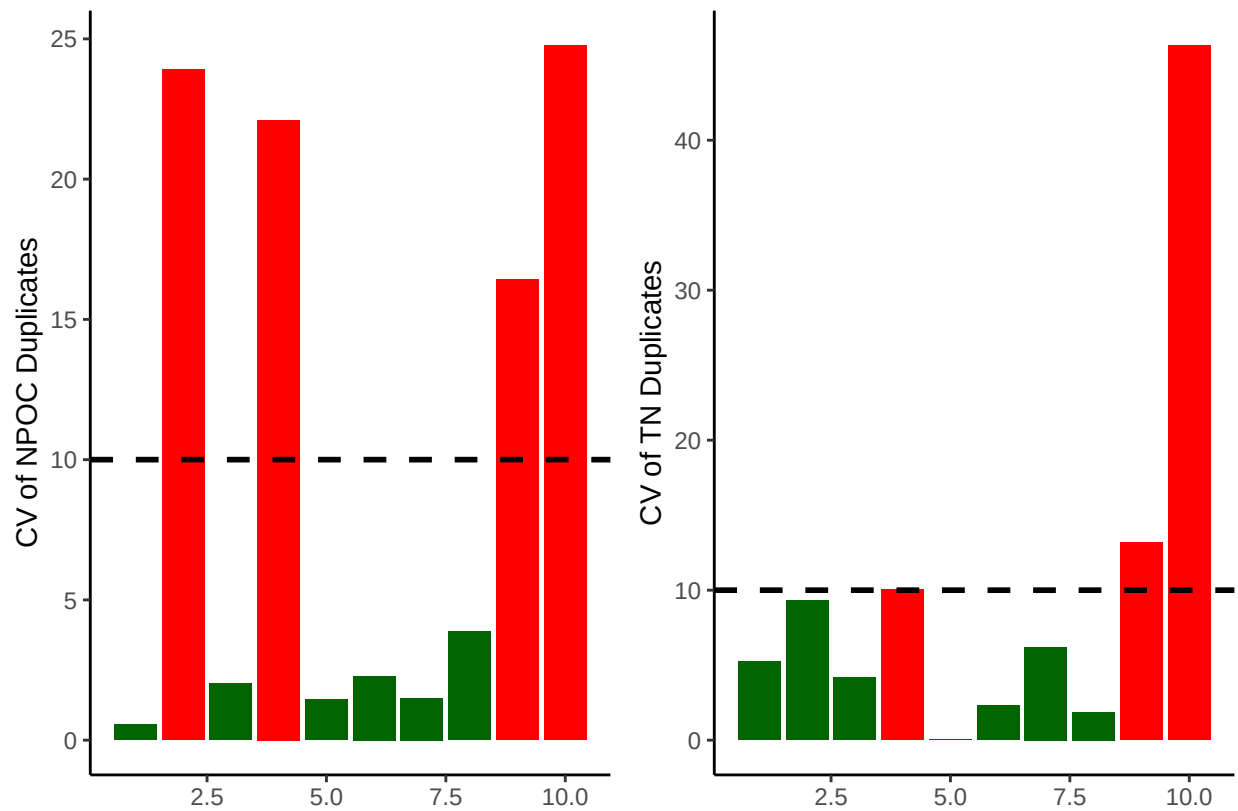
```
## nitrogen blanks:
```

```
## [1] -0.0001480769
```

Assess Duplicates

```
## Assess Duplicates
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

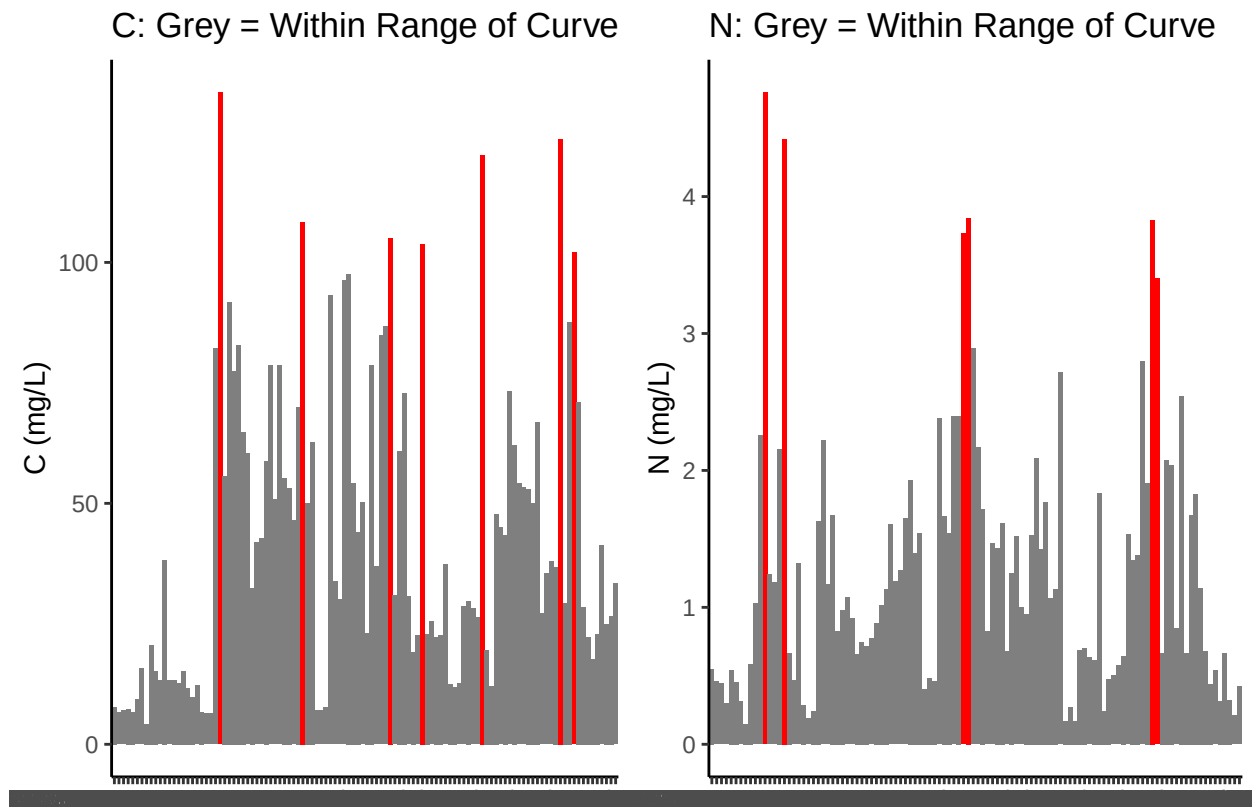


```
## [1] ">60% of Carbon Duplicates have a CV <10%"
```

```
## [1] ">60% of Nitrogen Duplicates have a CV <10%"
```

Sample Flagging - Are samples Within the range of the curve?

```
## Sample Flagging
```

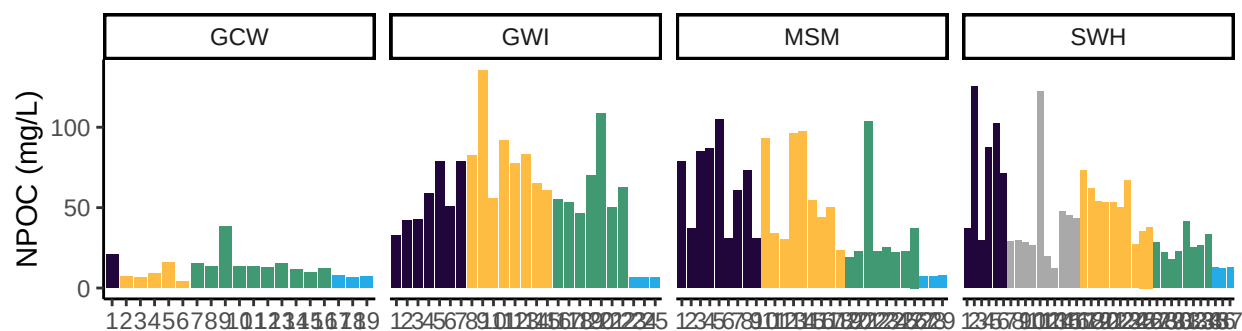


Visualize Data by Plot

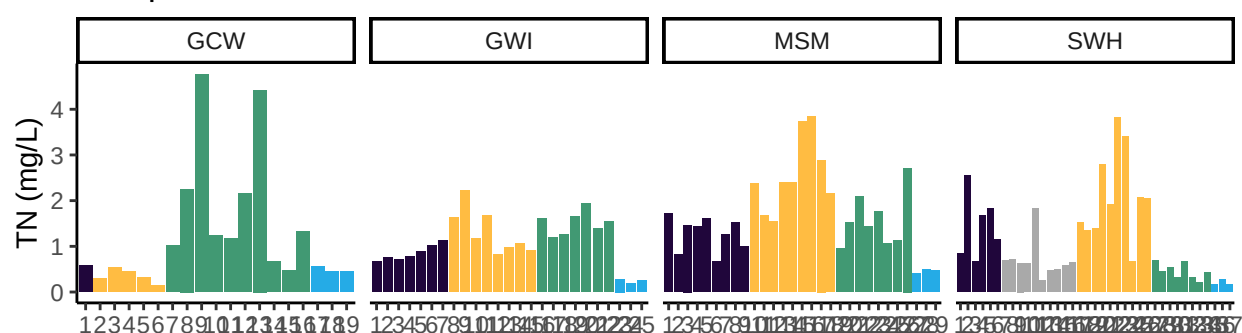
```
## Visualize Data
```

```
## Warning in rbind(c("202507", "GCW", "UP", "LysB", "20cm"), c("202507", "GCW", :
## number of columns of result is not a multiple of vector length (arg 1)
```

Samples: NPOC



Samples: TN



Convert data from mg/L to uMoles/L

Check to see if samples run match metadata & merge info

```
## Check Sample IDs with Metadata
```

```
## All sample IDs are present in metadata.
```

Export Processed Data

```
## Export Processed Data
```

```
## # A tibble: 6 x 21
```

```
##   Project      Region Site  Zone  Replicate Depth_cm Sample_ID  Year Month  Day
##   <chr>        <chr> <chr> <chr> <chr>      <int> <chr>    <int> <int> <int>
## 1 COMPASS: Sy~ CB    GCW  UP    B          20 202507_G~ 2025  7    16
## 2 COMPASS: Sy~ CB    GCW  TR    A          20 202507_G~ 2025  7    16
## 3 COMPASS: Sy~ CB    GCW  TR    B          10 202507_G~ 2025  7    16
## 4 COMPASS: Sy~ CB    GCW  TR    B          20 202507_G~ 2025  7    16
## 5 COMPASS: Sy~ CB    GCW  TR    C          10 202507_G~ 2025  7    16
## 6 COMPASS: Sy~ CB    GCW  TR    C          20 202507_G~ 2025  7    16
## # i 11 more variables: Time <chr>, Time_Zone <chr>, npoc_mgL <dbl>,
## #   npoc_uM <dbl>, npoc_flag <chr>, tdn_mgL <dbl>, tdn_uM <dbl>,
## #   tdn_flag <chr>, Analysis_runtime <chr>, Run_notes <chr>, Field_notes <chr>
```

#end